Customer Segmentation and Prediction

Michael Hansen, Keith John, Ujjawal Aggarwal, Daniel Chindris, Anay Misra, Sruthi Vadakuppa,
Anvesha Nain, Ishan Junnarkar, Nitya Jhaveri, Saumya Satishkumar Patel

INTRODUCTION

AgReliant, an agriculture company that specializes in corn and soybean seeds and sells throughout the midwest, aimed to improve their sales efficiency through data driven approach. Our team applied **machine learning techniques** in order to **analyze customer demographics** and **purchase history**, in order to **segment customers** and **predict** their **future buying** patterns.

STAGES OF OUR PROCESS

Stage One: Data Familiarization

Used **multiple datasets**, customer demographics, purchase history, and external market intelligence in order to focus on **data exploration** and **cleaning**. Leveraged **python** to clean the missing values and combine all datasets into a single, cleaned file

Stage Two: Model Development

Used the clean dataset, from stage one, to implement **various machine learning models** to **analyze customer behavior.** Used **clustering algorithms** for customer segmentation based on demographics and explored **tree-based models** such as **Random Forest** and **Gradient Boosting** for predicting future customer behavior. Lastly, used, **LightGBM** framework for customer prediction.

Stage Three: Model Visualization

To boost model interpretability, we used visualizations using **Python libraries** and **Tableau dashboards** in order to translate our model results to business insights.

RESULTS

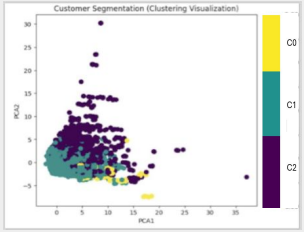
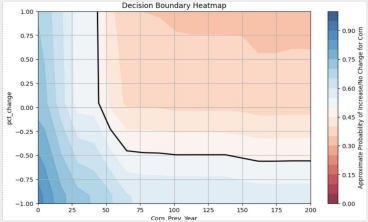
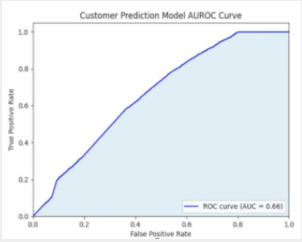


Image shows our clustering. It has a silhouette score of 0.5, which signifies that the clustering is reasonable. C0,C1 and C2 are the different clusters



Predicted likelihoods that customers would increase their purchase volume in the year 2024 with 65% accuracy.

Red: Customer more likely to decrease
Blue: Customer more likely to increase



The AUROC curve illustrates how well the model distinguishes between buyers and non-buyers. It was modeled using Random Forest.



Created a map of how well the customer prediction model performed when predicting the 2024 sales year in Tableau

Red counties: customer behavior with low accuracy

Green counties: customer behavior with high accuracy

MACHINE LEARNING MODELS

Customer Segmentation

BIRCH: a hierarchical clustering approach that incrementally builds a tree structure to summarize the dataset, enabling efficient clustering of large volumes of data while effectively managing noise and outliers.

Silhouette Score: a metric used to evaluate the quality of clustering, measuring how well each data point fits within its assigned cluster compared to other clusters, with values ranging from -I to 1 (We achieved a score of 0.60- excellent clustering)

Customer Prediction

RandomForest: Uses combinations of decision trees for classification. Averages results from multiple trees to reduce for overfitting and variance

LightGBM: Builds on trees sequentially, with each tree building on the errors of the last one allowing for high-speed results.

Feature Engineering: Created features taking aggregated findings from previous years and turning them into meaningful statistics to predict future years

CONCLUSION/FUTURE GOALS/ACKNOWLEDGEMENTS

We identified three customer segments using the BIRCH model. Our customer prediction model is under evaluation for accuracy and reliability. For sales forecasting, the Random Forest model achieved 65% accuracy. **Future improvements** will focus on feature engineering, hyperparameter tuning, and exploring additional algorithms.

Acknowledgements: Patrick Sadler. Jason Chia. The Data Mine Staff. AgReliant Staff for supporting this project

References: sci-kit learn. numpy. pandas. python documentation